

INTEREST RATE MODELS

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Interest rates: compounding

Investment horizon one year. Payment is in one year

- Annual compounding $(1 + i/1)^1$
- Semi annual compounding $(1 + i/2)^2$
- Quarterly compounding $(1 + i/4)^4$
- Monthly compounding $(1 + i/12)^{12}$
- Daily compounding $(1 + i/365)^{365}$
- n compounding $(1 + i/n)^n$

Interest rates: compounding

- Continuous compounding

$$\lim_{n \rightarrow \infty} (1 + i/n)^n = e^i$$

Interest rates: compounding

Investment horizon 2 years. Payment is in 2 years

- Bi-annual compounding $(1 + 2i/1)^1$
- Annual compounding $(1 + 2i/2)^2$
- Semi annual compounding $(1 + 2i/4)^4$
- Quarterly compounding $(1 + 2i/8)^8$
- Monthly compounding $(1 + 2i/24)^{24}$
- Daily compounding $(1 + 2i/730)^{730}$
- Continuous compounding $\lim_{n \rightarrow \infty} (1 + 2i/n)^n = e^{2i}$

Interest rates: conventions

- Trading days in a year (252)
- Holiday conventions. Calendars between countries differ
- Investment horizon one year. Payment is in one year
- Monthly compounding $(1 + i/24)^{12}$
- Monthly compounding $(1 + i_{31}/360)(1 + i_{28}/360)\dots(1 + i_{31}/360)$
- Monthly compounding $(1 + i_{30}/360)(1 + i_{30}/360)\dots(1 + i_{30}/360)$

Conventional rates (y) vs zero-coupon rates (i)

Suppose there are three bonds.

- Maturity 1 year, pays 1 in one year (no coupons). Price: $P_t^{(1)}$
 - ▶ $P_t^{(1)} = \frac{1}{1+i_{t,1}}$
- Maturity 2 years, pays coupon c in one year, 1 in two years. Price: $B_t^{(2)}$
 - ▶ The convention is $i_1 = y_1$. This is $P_t^{(1)} = B_t^{(1)}$
 - ▶ $B_t^{(2)} = \frac{c}{(1+y_{t,2})} + \frac{1}{(1+y_{t,2})^2}$
 - ▶ We note that it is the same interest rate discounting each payment
- Maturity 2 years, pays 1 in two years (no coupons). Price: $P_t^{(2)}$
 - ▶ What would the interest rate be of the bond with price $P_t^{(2)}$?
 - ▶ $P_t^{(2)} = \frac{1}{(1+i_{t,2})^2}$

Conventional rates (y) vs zero-coupon rates (i)

- $B_t^{(2)} = \frac{c}{(1+y_{t,2})} + \frac{1}{(1+y_{t,2})^2}$
- We have that $B_t^{(2)}$ should satisfy $B_t^{(2)} = \frac{c}{(1+i_{t,1})} + \frac{1}{(1+i_{t,2})^2}$
- Empirical problem: we know $B_t^{(2)}$ and $i_{t,1}$, how do you find $i_{t,2}$?
- This is a non linear equation

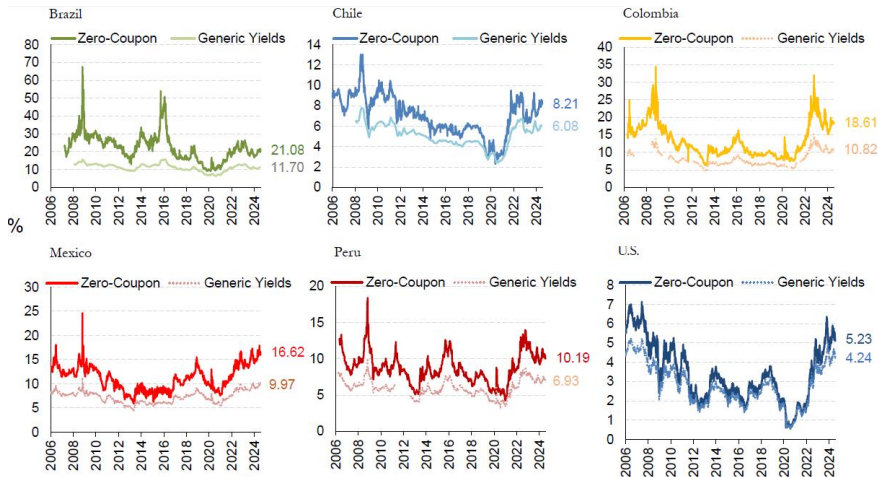
- Furthermore, suppose there is another government bond that pays a coupon c in the first year, a second coupon c in the second year and 1 in the third year.
- $B_t^{(3)} = \frac{c}{(1+y_{t,3})} + \frac{c}{(1+y_{t,3})^2} + \frac{1}{(1+y_{t,3})^3}$
- Iteratively, one obtains $i_{t,3}$ by solving the following equation
- $B_t^{(3)} = \frac{c}{(1+i_{t,1})} + \frac{c}{(1+i_{t,2})^2} + \frac{1}{(1+i_{t,3})^3}$

Conventional rates (y) vs zero-coupon rates (i)

- Given a set of $B_t^{(1)}, B_t^{(2)}, \dots, B_t^{(n)}$ and a coupons' structure, and $\{(n, y_{t,n})\}$
- One can then obtain $\{(n, i_{t,n})\}$
- The procedure is known as bootstrap (nothing to do with the econometric re-sampling)
- The general idea is that $\{(n, y_{t,n})\}$ and $\{(n, i_{t,n})\}$ have the same pricing information. Provided that the coupons are known, one can one into the other
- $\{(n, y_{t,n})\}$ are known as interest rates or yields
- $\{(n, i_{t,n})\}$ are known as zero coupon interest rates

Conventional rates (y) vs zero-coupon rates (i)

Figure 1: 10y Zero-coupon rates and 10y generic yields



Source: Bloomberg.

Term structure of interest rates

- The sets $\{(n, y_{t,n})\}$ and $\{(n, i_{t,n})\}$ are known as the term structure of interest rates or **yield curve**
- Modelling the term structure of interest rates
- Some intuition on the slope of the term structure of interest rates
 - ▶ Interest rates reflect the expectations of investors about the future economic situation; thus, a steepening of the yield curve reflects the market's expectation of an acceleration of inflation
- Three approaches:
 - 1 Statistic: principal component analysis (PCA)
 - 2 Parametric: Nelson-Siegel (NS)
 - 3 Financial Model: Affine term structure models (ATSM)

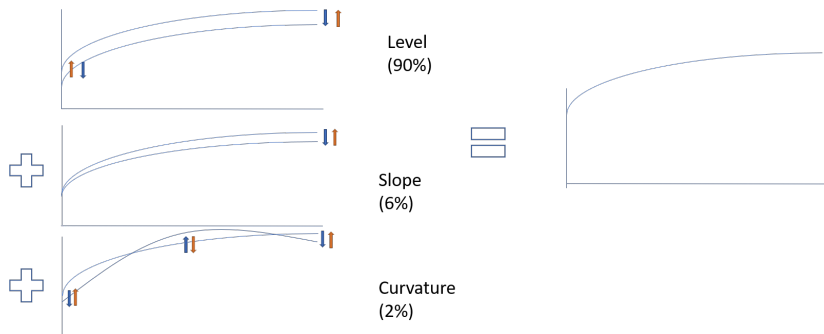
Principal component analysis (PCA)

- The sets $\{(n, y_{t,n})\}$ and $\{(n, i_{t,n})\}$ are known as the term structure of interest rates or yield curve
- Decompose $[y_{t,1}, y_{t,2}, \dots, y_{t,N}]$ using principal component analysis
- Nothing stops you for using either $\{(n, y_{t,n})\}$ and $\{(n, i_{t,n})\}$
- $i_{t,n} \approx \mu_n + \alpha_{n,1}f_{t,1} + \alpha_{n,2}f_{t,2} + \alpha_{n,3}f_{t,3}$
- $i_{t,n} \approx \mu_n + \alpha_{n,1}level_{t,1} + \alpha_{n,2}slope_{t,2} + \alpha_{n,3}curvature_{t,3}$

Principal component analysis (PCA)

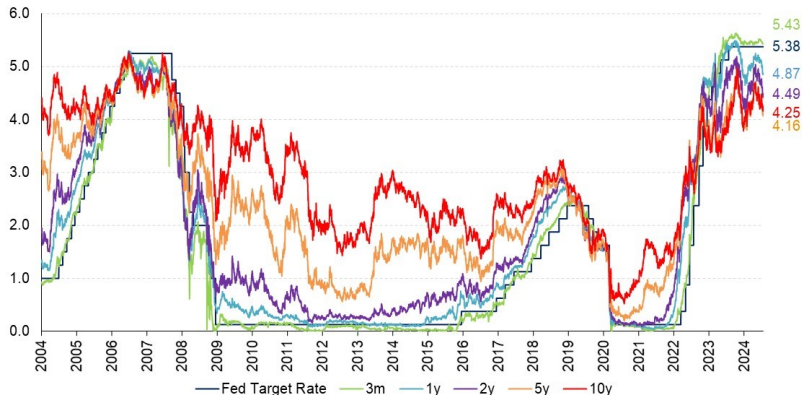
$$i_{t,n} \approx \mu_n + \alpha_{n,1} \text{level}_{t,1} + \alpha_{n,2} \text{slope}_{t,2} + \alpha_{n,3} \text{curvature}_{t,3}$$

Figure 2: PCA



Term structure of interest rates

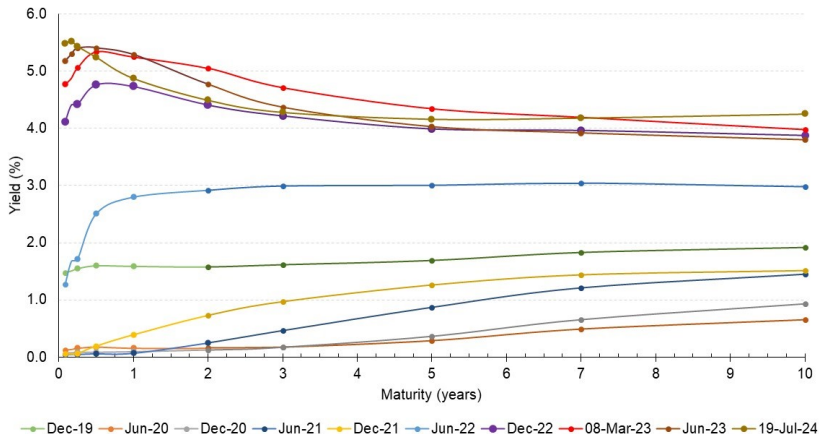
Figure 3: U.S. Treasury Yield Curve Rates



Source: U.S. Department of the Treasury.

Term structure of interest rates

Figure 4: U.S. Treasury Yield Curves



Source: U.S. Department of the Treasury.

Parametric: Nelson-Siegel (NS)

- **Nelson-Siegel** is a popular parametric model for the yield curve
- $$y(\tau) = \beta_0 + \beta_1 \left(\frac{1 - \exp(-\lambda\tau)}{\lambda\tau} \right) + \beta_2 \left(\frac{1 - \exp(-\lambda\tau)}{\lambda\tau} - \exp(-\lambda\tau) \right)$$
- where $y(\tau)$ represents the zero-coupon yield curve function with maturity τ . The parameters are $\beta = (\beta_0, \beta_1, \beta_2)$, λ , τ maturity
- Note that β_0 refers to the level, β_1 to the **slope** and β_2 to **the curvature** of the yield curve
- Expressions in parenthesis are called factor loadings which are defined by maturity τ and factor λ .
- That equation can be written as a regression equation
- $$y_t(\tau) = H\beta_t + \epsilon_t,$$

Parametric: Nelson-Siegel (NS)

- where $y_t(\tau)$ represents a vector of yields observed in the moment t for T maturities in the vector τ , and H represents factor loadings matrix

$$H = \begin{bmatrix} 1 & \frac{1 - \exp(-\lambda\tau_1)}{\lambda\tau_1} & \frac{1 - \exp(-\lambda\tau_1)}{\lambda\tau_1} - \exp(-\lambda\tau_1) \\ 1 & \frac{1 - \exp(-\lambda\tau_2)}{\lambda\tau_2} & \frac{1 - \exp(-\lambda\tau_2)}{\lambda\tau_2} - \exp(-\lambda\tau_2) \\ \vdots & \vdots & \vdots \\ 1 & \frac{1 - \exp(-\lambda\tau_T)}{\lambda\tau_T} & \frac{1 - \exp(-\lambda\tau_T)}{\lambda\tau_T} - \exp(-\lambda\tau_T) \end{bmatrix}$$

Affine term structure models (ATSM)

- Models in which interest rate are a linear function of risk factors
- Affine models are popular because they are tractable and relatively easy to est

Three key components

- VAR capturing the risk or macro factors
- A linear model for the interest rates, in particular for the short-term rate
- A model for the stochastic discount factor (SDF)

Affine term structure models (ATSM): VAR

- Consider the VAR with the ATSM
- $X_{t+1} = \mu + Ax_t + \Sigma\epsilon_{t+1}$ where ϵ_{t+1} distributes $N(0, I)$
- There are several ways to go about the estimation of a ATSM
 - 1 Extract from the interest rates time series, for example, PCA
 - 2 Use observable financial time series, FX, VIX, etc.
 - 3 Use latent variables obtained within the estimation of the ATSM

Decomposing the 10-year zero-coupon interest rate to expected inflation

- Generalized Fisher Equation

$$(1 + 10 \cdot i_t^{(10)}) = (1 + r_t)^{10} (1 + E_t(\pi_t^{10}))^{10} (1 + 10 \cdot TP_t^{(10)})$$

- $i_t^{(10)}$ is the zero-coupon 10-year interest rate in simple compounding
- r_t is the 10-year real interest rate
- $E_t(\pi_t^{10})$ is the 10-year expected inflation
- $TP_t^{(10)}$ is the 10-year term premium in simple compounding
- The term premium is defined as the compensation that investors require for bearing the risk that interest rates may change over the life of the bond

Decomposing the 10-year zero-coupon interest rate to expected inflation

We know $i_t^{(10)}$ and $TP_t^{(10)}$, and we use the natural (Wicksellian) rate for r_t . Thus,

$$\left(1 + E_t(\pi_t^{(10)})\right)^{10} = \frac{(1 + 10 \cdot i_t^{(10)})}{(1 + r_t)^{10}(1 + 10 \cdot TP_t^{(10)})}$$
$$E_t(\pi_t^{(10)}) = \exp\left(\frac{1}{10} \ln \frac{(1 + 10 \cdot i_t^{(10)})}{(1 + r_t)^{10}(1 + 10 \cdot TP_t^{(10)})}\right) - 1$$

Decomposing the 10-year zero-coupon interest rate to expected inflation

Figure 5: Decomposing the 10-year zero-coupon to expected inflation (Average – Jan. 2015 to July 19, 2024)

Country	$i^{(10)}$	$TP^{(10)}$	Natural (Wicksellian) Real Interest Rate			Expected Inflation		Inflation Target
			Low	High		High	Low	
Brazil	19.68	1.66	4.00	5.00	¹	5.58	4.57	3.98*
Chile	5.85	0.59	0.60	1.50	²	3.49	2.58	3.00
Colombia	12.57	2.89	1.40	2.40	³	4.30	3.29	3.00
Mexico	11.28	0.28	1.80	2.60	⁴	5.65	4.82	3.00
Peru	8.56	2.73	0.50	1.50	⁵	3.33	2.31	2.00
U.S.	2.77	-0.41	1.42	2.23	⁶	1.47	0.67	2.00

Source: Own estimations with data from Bloomberg for the interest rates and Bloomberg and FRBNY for the term premia.

Decomposing the 10-year zero-coupon interest rate to risk neutral

Take as the long-term rate the 10-year zero-coupon interest rate (in simple composition), as the short-term rate the 1-month interest rate, and the term premium (in simple composition).

$$(1 + 10 \cdot i_t^{(10)}) = \left[\left(1 + \frac{E_t i_t^{1m}}{12} \right) \dots \left(1 + \frac{E_t i_{t+119m}^{1m}}{12} \right) \right] (1 + 10 \cdot TP_t^{(10)})$$

The risk-neutral interest rate is given by:

$$(1 + 10 \cdot i_t^{(10,*)}) \equiv \left[\left(1 + \frac{E_t i_t^{1m}}{12} \right) \dots \left(1 + \frac{E_t i_{t+119m}^{1m}}{12} \right) \right]$$

Hence:

$$(1 + 10 \cdot i_t^{(10)}) = (1 + 10 \cdot i_t^{(10,*)}) (1 + 10 \cdot TP_t^{(10)})$$

Decomposing the 10-year zero-coupon interest rate to risk neutral

$$\text{Interest Rate} = (1 + \text{Risk-Neutral})(1 + \text{Term Premium})$$

Figure 6: Decomposing the 10-year zero-coupon to risk neutral

		$i^{(10)}$	$i^{(10,*)}$	Term Premium
Brazil	Average 2022-2024	21.02	16.21	1.84
	Last Observation	21.08	15.86	2.02
Chile	Average 2022-2024	7.72	6.15	0.98
	Last Observation	8.21	6.05	1.35
Colombia	Average 2022-2024	19.41	8.65	5.77
	Last Observation	18.61	7.54	6.30
Mexico	Average 2022-2024	14.59	14.66	-0.03
	Last Observation	16.62	15.29	0.53
Peru	Average 2022-2024	10.90	5.10	3.84
	Last Observation	10.19	5.00	3.47
U.S.	Average 2022-2024	4.36	5.03	-0.45
	Last Observation	5.23	5.43	-0.13

Last observation: July 19, 2024. Source: Own Estimations with data from Bloomberg.

Real term structure (inflation risk)

The Fisher equation defines the real interest rate as:

$$i_t = r_t + E_t(\pi_{t+1})$$

- i_t = nominal, one - period interest rate
- r_t = real interest rate
- $E_t(\pi_{t+1})$ = expected inflation rate in $t + 1$

This equation isn't quite right. There's a cross-product term and an inflation risk term that are omitted.

Real term structure (inflation risk)

- $P_t \equiv$ price of goods in units of USD
- $1/P_t$ is the price of USD in units of goods
- Inflation is the rate of change in the price of goods, denoted π_{t+1} :

$$\pi_{t+1} \equiv \frac{P_{t+1}}{P_t} - 1$$

or

$$\pi_{t+1} \equiv \ln\left(\frac{P_{t+1}}{P_t}\right)$$

Real term structure (inflation risk)

Real return:

$$\text{Real return} = \frac{\text{Number of Goods Received}}{\text{Number of Goods Paid}}$$

Real return on a nominal bond

- A nominal bond costs USD $B_t^{(1)}$
- Equivalently, in units of goods, it costs:

$$\text{Goods } \frac{B_t^{(1)}}{P_t}$$

- A nominal bond pays USD 1.0. Equivalently, in units of goods, it pays:

$$\text{Goods } \frac{1.0}{P_t}$$

Real term structure (inflation risk)

The real return is therefore

$$\text{Real return} = \frac{P_t}{B_t^{(1)} P_{t+1}}$$

Therefore, we can write the real return on the nominal bond as r_{t+1} :

$$(1 + r_{t+1}) = \frac{(1 + i_{t+1})}{(1 + \pi_{t+1})}$$

Note that, because of inflation, this return is random.

References

- CEMLA (2023). *CEMLA Governors' Presentation*.
- CEMLA (2022, 2021). *Course on Financial Econometrics*.
- Telmer, C. (2013b). Modeling the Real Term Structure. Carnegie Mellon University.